

Sreya Sarkar

Curriculum Vitae

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Education

- 2023 - Aug 2026 **Ph.D., Statistics**, The University of Iowa (UI), Iowa City, USA.
Advisor: [Dr. Sanvesh Srivastava](#)
GPA: 3.97/4
- 2021 - 2023 **M.S., Statistics**, The University of Iowa, Iowa City, USA.
Advisor: [Dr. Sanvesh Srivastava](#)
GPA: 3.73/4
- 2019 - 2021 **M.Sc., Statistics**, Indian Institute of Technology (IIT), Kanpur, India.
Advisor: [Dr. Minerva Mukhopadhyay](#)
GPA: 8.5/10
- 2016 - 2019 **B.Sc. (Honours), Statistics**, St. Xavier's College (SXC), Kolkata, India.
Advisor: [Dr. Surupa Chakraborty](#)
GPA: 8.6/10

Research Interests

High-dimensional repeated-measures data, Multimodal data, Tensor Regression, Bayesian Statistics, Computational Statistics.

Experience

- May 2023– May 2025 **Graduate Research Assistant**, *DEPARTMENT OF STATISTICS, THE UNIVERSITY OF IOWA*,
Supervisor: [Dr. Sanvesh Srivastava](#).
- Developed a computationally efficient Bayesian methodology for prediction and fixed effects selection with mixed-effects models in high-dimensional repeated-measures data with tensor-valued covariates.
 - Leveraged random projections to handle high-dimensional random effects covariance matrices and established theoretical guarantees of the associated Bayes risk of the proposed method.
 - Demonstrated the advantages of the proposed approach over existing penalized quasi-likelihood methods through extensive simulation studies and real-data analyses in biomedical and behavioral science applications.
- Aug 2021– Ongoing **Graduate Teaching Assistant**, *DEPARTMENT OF STATISTICS, THE UNIVERSITY OF IOWA*.
- Graduate Student Instructor for Elementary Statistics and Inference (STAT 1020)
 - Teaching Assistant and/or Grader for following courses:
 - Bayesian Statistics (STAT 4520)
 - Statistical Learning (STAT 4540)
 - Probabilistic Statistical Learning (DATA 4750)
 - Probability and Statistics for Engineering (STAT 2020)
 - Statistics for Business (STAT 1030)

Summer 2024 **Research Assistant**, *DEPARTMENT OF GASTROENTEROLOGY AND HEPATOLOGY, WEILL CORNELL MEDICINE*,

Supervisor: [Dr. Himel Mallick](#).

- Developed a Bayesian cooperative learning method for multimodal integration, utilizing intermediate fusion and Bayesian regularization for continuous, binary, and survival outcomes.
- Developed a meta-network analysis method to integrate multiple microbiome studies, using concepts from multi-study machine learning and differential network analysis.
- Conducted comprehensive benchmarking of these methods using realistic synthetic data and applied them to real-world microbiome and multi-omics datasets, including pregnancy outcomes and cancer immunotherapy response.

Publications/Preprints

- In Submission

- 2026 **Sarkar, S.**, Khare, K., Srivastava, S., CoMET: A Compressed Bayesian Mixed-Effects Model for High-Dimensional Tensors, [arXiv](#). (*Major Revision requested from Bayesian Analysis*)
- 2026 Roy, S., **Sarkar, S.**, Banerjee C., Paul E., Basak P., Mallick, H., Conformalized Multiview Learning, [bioRxiv](#). (*Major Revision requested from Statistical Applications in Genetics and Molecular Biology*)
- 2025 Roy, S.*, **Sarkar, S.***, Paul E., Basak P., Yi N., Mallick, H., Bayesian Cooperative Learning for Multimodal Integration, [bioRxiv](#) (*Major Revision requested from Journal of Computational and Graphical Statistics*) * These authors contributed equally.
- 2025 **Sarkar, S.**, Khare, K., Srivastava, S., Bayesian Compressed Mixed-Effects Models, [arXiv](#). (*Revision requested from Journal of Machine Learning Research*)

- In Preparation

Sarkar, S., Roy, S., Zhang, X., Rahanavard, A., Mallick, H., Multi-study Meta-Analysis of Differential Microbiome Networks.

Sarkar, S., Khare, K., Srivastava, S., Generalized Compressed Bayesian Mixed-Effects Model for High-Dimensional Tensors.

R Software

- Developer ◦ [BayesCoMET](#), Compressed Bayesian Mixed-Models for Tensors.
- [BayesCOOP](#), Bayesian Cooperative Learning.
- [Coracle](#), Conformalized Multiview Learning.
- [cme](#), Bayesian Compressed Mixed-Effects Models.
- [muffinette](#), Multi-study Meta-network Analysis.

Conferences & Workshops

- Aug 2025 **Invited Talk**, on “*Bayesian Cooperative Learning for Multimodal Integration*”, Joint Statistical Meetings (JSM) 2025, Nashville, USA.
- Aug 2025 **Contributed Talk**, on “*Bayesian Compressed Mixed-Effects Models*”, Joint Statistical Meetings (JSM) 2025, Nashville, USA.
- Apr 2025 **Lightning Talk**, on “*Bayesian Compressed Mixed-Effects Models*”, Bayesian Young Statisticians Meeting (BAYSM) 2025, j-ISBA, Virtual.
- Dec 2024 **Poster Presentation**, on “*Bayesian Compressed Mixed-Effects Models*”, International Indian Statistical Association (IISA) Conference 2024, Kochi, India.
- Dec 2024 **Omics Data Science Workshop**, IISA Conference 2024, Kochi, India.

Awards

- Aug 2025 **UI Graduate Student Senate Travel Award**, - *JSM 2025, Nashville, USA*.
Dec 2024 **NSF Travel Award**, - *IISA Conference 2024, Kochi, India*.
Jul 2019 **Fr. Lafont Scholarship for Academic Performance**, - *SXC Kolkata, India*.
July 2016 - **INSPIRE Scholarship, Department of Science and Technology**, - *Government of India*.
July 2021

Professional Services

- Sep 2024-Present **Reviewer**, *Electronic Journal of Statistics, Sankhya B, STAT*.
Aug 2025 **Session Chair**, *Invited Session on “Advances in Foundation Models and LLMs for Biomedical Data Science”, JSM 2025, Nashville, USA*.
Dec 2024 **Session Chair**, *Panel Discussion on “Breaking Down Barriers and Cultivating a Culture of Collaboration and Leadership in Statistical Sciences: Enhancing Synergy for Collective Growth”, IISA Conference 2024, Kochi, India*.
Apr 2024 - Present **Treasurer**, *Statistics Student Organization*, Department of Statistics and Actuarial Science, The University of Iowa.

References

1. **Sanvesh Srivastava (Ph.D. Advisor)**
Associate Professor
Department of Statistics and Actuarial Science
The University of Iowa
sanvesh-srivastava@uiowa.edu
2. **Kshitij Khare (Collaborator)**
Professor
Department of Statistics
University of Florida
kdkhare@stat.ufl.edu
3. **Himel Mallick (Collaborator)**
Assistant Professor
Division of Biostatistics
Department of Population Health Sciences
Weill Cornell Medicine
him4004@med.cornell.edu